NAWSHIN ALAM

Environmental Oceanography — Ecosystem Modelling — Remote Sensing & GIS

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Education

B.Sc. in Oceanography **2020–2025**
Bangladesh Maritime University (BMU) *Bangladesh*

- CGPA: **3.74/4.00**.
- Position: 7th

Research Experience

Undergraduate Research Project: Evaluation of Ecosystem Health of the Saint Martin's Island Marine Protected Area of Bangladesh

Integrated Approach for Strategic Mangrove Restoration Plan

Funded by: United Nations Development Programme (UNDP)
Awarded in: Youth4Climate 2025
Position: Project Lead
Duration: January 2026–December 2027 (Approx.)

Understanding the Spatiotemporal Dynamics and Complexity of Urban Heat Islands Using Remote Sensing and Machine Learning

Funded by: Ministry of Science and Technology (MoST), Bangladesh
PD: Maisha Farzana Methila
Position: Research Associate
Duration: March 2026–June 2026

Assessing Ecosystem Health of a Deltaic Mangrove Island, Dubla Island, Sundarbans Mangrove Forest, through In Situ and Satellite Observation

Funded by: BMU
PI: Dr. S. M. Mustafizur Rahman
Position: Research Assistant
Duration: July 2024–Present

Assessment and Zoning of Marine Protected Areas for Biodiversity Conservation in Bangladesh: A Comparative Study of Existing MPAs and Identification of Potential Sites

Funded by: MoST, Bangladesh
PI: Dr. S. M. Mustafizur Rahman
Position: Research Assistant
Duration: July 2023–December 2024

Publications

1. Hasan, M. A., Koly, F., Sowrav, F. F., Rahman, S. M. M., **Alam, N.**, Masud-Ul-Alam, M., Zuhaib Kabir, M., Sarder, S. R., & Ahmed, K. (2026). Assessing the dynamics of water quality and their implications on biochemical health of the river ecosystem: An RS-based study of the Shitalakshya River, Dhaka. *Bangladesh Maritime Journal, 10*(1), 2520–1840. https://doi.org/10.70279/bmj-v2-1124

2. Rahman, S. M., Sowrav, S. F., Hasan, M. A., Koly, F. A., Hossain, K. A., Hoque, M. M., Nahar, N. U., **Alam, N.**, & Iqball, M. S. (2026). Preliminary assessment of river ecosystem status in a peri-urban area: A case study of the Lower Shitalakshya River, Bangladesh. *Water, 18*(20), 2586. https://doi.org/10.3390/w18202586

Technical Skills

Programming Language	Python, MATLAB, R, LaTeX, Ferret, SPSS
Software Skill	ArcGIS Pro, Google Earth Engine (GEE), MS Office, Canva, Adobe Illustrator, KoBoToolbox
Instruments	Spectrophotometer, CTD, Algal Torch, Multiparameter, Terrameter, LiDAR, RTK GPS, Atomic Absorption Spectroscopy (AAS)

Awards & Fellowships

- **National Science and Technology (NST) Fellowship**, MoST—Research on coastal productivity modelling.
- **Vice-Chancellor Scholarship**, Bangladesh Maritime University, 2025.

Selected Training & Certifications

- Workshop on **Coastal and Regional Ocean Community (CROCO) Model**, Bangladesh Maritime University, 2025.
- Training Course on **Numerical Model and Artificial Intelligence for Ocean Science (NMAI4OS)**, National Oceanographic and Maritime Institute.

Conferences & Scientific Events

- **25th International Conference of the North Indian Ocean Hydrographic Commission (NIOHC)**, Chattogram, Bangladesh, 10–12 May 2026.
- **Seabed Mapping: Enabling Ocean Action**, Dhaka, Bangladesh, 21 June 2025.
- **UN Decade of Ocean Science: Imperatives for Bangladesh 2021–2030**, Dhaka, Bangladesh, 17 November 2022.

Leadership & Service

Advisor — **May 2026–Present**
Maritime University Ocean Club — *BMU*

Communication Secretary — **Jul 2022–Jun 2023**
YouthMappers at BMU — *Bangladesh*

Professional References

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Dr. Mohan Kumar Das
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